RTD Denver EMU (50) Procurement Technical Specification of Ethernet Switch

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1. Introduction

1.1 General

This Procurement Technical Specification (PTS) defines the technical requirements for Ethernet Switch to be purchased by Hyundai Rotem Company and installed in the Electric Multiple Unit (EMU) Commuter Rail Cars, identified as the “Fastrack Eagle Project” cars, to be manufactured by Hyundai Rotem and to be operated by the DENVER Regional Transportation District (“Owner”). The cars will be operated by DENVER RTD over the DENVER Metropolitan Area Railroad (East Corridor, Gold Line, and Northwest Rail).

1.2 General Requirements

It is the intent that all Suppliers providing equipment or services to Hyundai Rotem shall comply with the Customer’s requirements. When the Customer does not have any specific requirements, the Supplier shall comply with the requirements of this PTS as appropriate. Suppliers must comply with the requirements stated herein unless otherwise agreed to in writing by Hyundai Rotem. Any conflict between the Customer’s requirements and this specification shall be brought to Hyundai Rotem’s immediate attention for resolution. Notwithstanding conflict resolution, the most restrictive requirement shall apply. The term “Customer”, as used in this document, refers to Hyundai Rotem’s Customer (DENVER RTD).

This PTS in no way relieves the Supplier from any requirements specified in the Technical Specification.

All formal communication between Hyundai Rotem staff and Suppliers must be through the Procurement function at Hyundai Rotem. Formal transmittal from Suppliers of drawings, procedures, reports, and analyses referenced in this document must be sent to the responsible Hyundai Rotem Procurement Representative defined in the instruction letter transmitting this PTS. Formal approvals and documentation from Hyundai Rotem to Suppliers will be sent via the responsible Procurement Representative.

The order of precedence for Supplier technical requirements shall be as follows:

1) The Purchase Order between Hyundai Rotem and the Supplier

2) The Contract between Hyundai Rotem and the Customer

3) Hyundai Rotem’s PTS

4) DENVER RTD Technical Specification Contract

As a guide, the sections of the aforementioned documents are considered to contain specific and relevant requirements have been referenced in this document. However, this in no way diminishes the Suppliers’ responsibility for compliance with the Technical Specification. The complete requirements are those found within this PTS document, the Technical Specification, the Contract, addendums, and the documents referenced therein. It shall be the Supplier's responsibility to assure that current revisions of the subject documents are being utilized, and to assure that equipment, documentation, and services furnished pursuant to this PTS are in full compliance with all applicable documents.

1.3 Train Configuration

The configuration of on unit is as follows:

(a) Total number of train; 25 train sets.

(b) Each train(Unit) will consists of MA and MB.

2. Definitions and Abbreviations

This section includes a list of all definitions and abbreviations applicable to PTS, GS and TS.

2.1 Definitions

The following definitions are applicable to the PTS, Technical Specification for the Purchase of DENVER RTD (TS).

"**Contractor**" means the persons or person appointed by the Employer to undertake the execution of the works for Hyundai-Rotem. In order to avoid misunderstanding the roles of the Contractor in GS and TS, the term “Contractor” shall be read as “Subcontractor” for those clauses referred to in the GS and TS by the PTS.

"**Engineer**" means the personnel appointed by DENVER RTD for the purpose of the DENVER RTD Electrical Multipule Unit.

“**Subcontractor**” means a supplier of complete Ethernet Switch to Hyundai-Rotem and Contractor specified in TS with regard to Ethernet Switch. Subcontractor shall carry out the works in accordance with TS with regard to Ethernet Switch.

2.2 Abbreviations

The following abbreviations shall apply to the PTS and TS.

GS: General Specification for the purchase of Electrical Multiple Unit Commuter Railcars.

TS: Technical Specification for the purchase of Electrical Multiple Unit Commuter Railcars.

PTS: Procurement Technical Specification for DENVER project issued by Hyundai-Rotem

RTD: Regional Transportation District

NA: Not applicable

EMI/EMC : Electro Magnetic Interference / Compatibility

S.A : System Assurance

RAMS : Reliability Availability Maintainability System safety

CDR : Conceptual Design Review

PDR : Preliminary Design Review

FDR : Final Design Review

3. Scope of supply and Work

3.1 Hardware

The Subcontractor shall be responsible for the provision of all equipments for the Ethernet Switch. The system shall be complied the DENVER RTD specification (CONTRACT PART I. 3.17.1) and the following specification or more. Design shall be based on the altitude conditions and climatic conditions of the Denver area. Equipment to be housed in equipment houses shall be rated for an operating temperature range of -40℉ ~ 160℉ at 0% to 95% relative humidity, non-condensing. Equipment shall be fulfilled with IEC61373, IEC60068-2-32, EN50155, EN50121-3-2, EN50121-4, EN61000-4-2, EN61000-4-3, EN61000-4-4, EN61000-4-5, EN61000-4-6, EN61000-4-8, EN61000-4-11 and EN61000-4-12. Subcontractor shall provide all components related to Ethernet Switch, but not limited to, the followings:

1) Managed Ethernet Switch
   - 1-1) Integrated Ethernet Switch unit
   - 1-2) Number of Port: Minimum 8 Ports (Proposal possible)
   - 1-3) Connectors (Incl. mating connectors for car side)
   - 1-4) Cable (Electric Power Cable, Data Cable and Special Cable)
   - 1-5) Pin and Sockets
   - 1-6) Bolts, Nuts and All Hardware for Installation
   - 1-7) Ethernet Redundancy Function
   - 1-8) Bypass Function
   - 1-9) Etc.
2) Jumper Cable & Plug Assembly for Ethernet Connection

3.2 Software

The Subcontractor shall provide, as a minimum, the followings:

1) Ethernet Switch shall be complied to the followings Standards:
   - 1-1) IEEE 802.3 for 10BaseT
   - 1-2) IEEE 802.3u for 100BaseT(X)
   - 1-3) IEEE 802.3ab for 1000BaseT(X)

1-4) IEEE 802.3x for Flow Control

1-5) IEEE 802.1D for Spanning Tree Protocol

1-6) IEEE 802.1w for Rapid STP

1-7) IEEE 802.1Q for VLAN Tagging

1-8) IEEE 802.1p for Class of Service

1-9) IEEE 802.1X for Authentication

1-10) IEEE 802.3ad for Port Trunk with LACP

2) And, Ethernet Switch shall be complied to the followings Protocols:

2-1) IGMPv1/v2

2-2) GMRP

2-3) GVRP

2-4) SNMPv1/v2C/v3

2-5) DHCP Server/Client

2-6) DHCP Option 66/67/82

2-7) BootP

2-8) TFTP

2-9) SNTP

2-10) SMTP

2-11) RARP

2-12) RMON

2-13) HTTP

2-14) HTTPS

2-15) Talent

2-16) SSH

2-17) Syslog

2-18) LLDP

2-19) IEEE 1588 PTP

2-20) Modbus/TCP

2-21) IPv6

3) And, Ethernet Switch shall be complied to the followings MIB:

3-1) MIB-II

3-2) Ethernet-like MIB

3-3) P-BRIDGE MIB

3-4) Q-BRIDGE MIB

3-5) Bridge MIB

3-6) RSTP MIB

3-7) RMON MIB Group 1, 2, 3, 9

4) Also, Ethernet Switch shall be complied to the followings Flow Control:

4-1) IEEE 802.3x Flow Control

4-2) Back Pressure Flow Control

5) Non-commercially available software

The subcontractor shall provide the all non-commercially available software defined in the TS.

6) Commercially Available software

All commercially available software shall be listed-up by the subcontractor.

3.3 Interfaces with other Carborne Equipment

The Subcontractor shall provide all hardware and engineering work for the interfaces with other Carborne Equipment, as a minimum, the following:

1) Communications System

Ethernet Switch shall be equipped with the function of interface between Controller of Communications System and Communication Management Unit (hereafter CMU) of Positive Train Control (hereafter PTC) System. Controller of Communications System has the function which can be communicated between Office Communication Center (hereafter OCC) and Carborne Intercoms by using WiMax, as VoIP. And, Controller of Communications System has the function which can be transmitted the GPS data from GPS antenna. GPS antennas will be connected the CMU of PTC System. Also, Controller of Communications System has the function which can be updated the contents and scenario of communications system to the sub-items of communications system at the CRMF by using Wi-Fi. Ethernet Switch shall be complied with the requirements of interface with Communications System.

2) CCTV System

Ethernet Switch shall be equipped with the function of interface between Digital Video Recorder (hereafter DVR) and CMU of PTC System. DVR has the function which can be transmitted the live video data of IP-based CCTV camera to OCC by using WiMax. Also, DVR has the function which can be downloaded the recorded video data of DVR to Workstation at CRMF by using Wi-Fi. Ethernet Switch shall be complied with the requirements of interface with CCTV System.

3) PTC System

Ethernet Switch shall be equipped with the function of interface with CMU of PTC System. CMU will be connected to the antennas. (WiMax, Wi-Fi and GPS) Ethernet Switch shall be complied with the requirements of interface with PTC System.

3.4 Spares, Special Tools and Testing Equipment

The Contractor shall furnish the following capital spares, test equipment, and special tools. If a Rolling Stock Option is exercised, the proportional number of capital spares will be provided. Selected special tools and test equipment shall also be required depending on the nature of the tools/equipment and its intended usage.

3.4.1 Special Tools and Testing Equipment

All tools, excluding tools commonly available from standard suppliers such as Granger and/or Snap-On that are required or desirable for the effective/efficient conduct of maintenance and repair of the vehicles will be provided as a “special tool” in quantities necessary to support the level and intensity of vehicle maintenance anticipated. This requirement extends through mechanical, electrical, electromechanical, and electronic tools, including software products. These tools will be provided within 60 days of the delivery of the first production Car of the Base Order.

All test equipment required or desirable for the conduct of maintenance and repair activities of the Car will be provided in sufficient quantity to adequately support the maintenance operation. All such test equipment to be provided by the Contractor regardless of the equipment’s availability from other common sources. Portable test equipment (Two of each kind, including instruction manuals, cables, connectors and associated software), shall be provided for all on-board systems. All portable test equipment shall be delivered within 30 days of the delivery of the first production Car of the Base Order. One set of bench test equipment shall be provided for the purpose of testing, troubleshooting and calibrating all electrical, electronic, mechanical and electro-mechanical components of each vehicle subsystem. All test equipment, including bench test equipment, shall be delivered within 90 days of the delivery of the first production Car of the Base Order.

The Subcontractor shall provide, at a minimum, the following:

The spare parts shall be based on the principal that local maintenance will be to Level One. This level assumes the diagnosis and replacement of major parts and line replaceable units (LRU) locally. Defective units and/or their sub-assemblies will be returned to the subcontractor for repair.

The Subcontractor shall recommend and submit the detailed list of spare parts for maintenance.

The Subcontractor may recommend additional spare parts and separately quote spare parts if he esteems necessary for maintenance and repair of the supplied Ethernet Switch for Hyundai Rotem’s further consideration as separate purchase.

The Subcontractor shall propose the spare parts according to Attachment of Contract Part I as a minimum.

3.4.2 Spare

A minimum of 2 Car sets of all replaceable/rebuildable interior body units will be provided as capital spares deliverable to the Site before the delivery of the last production Car of the Base Order. This provision will include but not be limited to all interior body panels and fixtures, seats, window assemblies, windshields assemblies door leafs, end doors, and other body parts as may be damaged and require replacement.

A minimum of 3 Car sets of all replaceable/rebuildable mechanical/electrical units will be provided as capital spares deliverable to the Site before the delivery of the last production Car of the Base Order. This provision will include but not be limited to fully dressed (built-up, ready to install) trucks, traction motors, air compressors, air conditioning compressors, air conditioning condensers, complete air conditioning units (including frame), propulsion equipment, auxiliary power equipment, train control equipment, communications/radio equipment, safety equipment, master controller, operators "desk" equipment, door operators and other major mechanical, electrical, and electromechanical equipment.

A minimum of 4 Car sets of couplers and draw bar assemblies will be provided as capital spares deliverable to the Site before the delivery of the last production Car of the Base Order. In addition, 4 compromise couplers will be provided.

The subcontractor must supply all the accessories including but not limited to spares, special tools, test equipments and etc.

3.5 Interface Responsibilities

The subcontractor shall provide recommendations and specifications on the installation of all the subcontractor's supply equipment. The subcontractor shall provide the interface specification between Ethernet Switch and vehicle equipments (as TS, but not limited to) and performed the interface test between Ethernet Switch and vehicle equipments. Hyundai Rotem will perform the installing design and installing work of Ethernet Switch with the subcontractor's information. The subcontractor shall develop and provide performance criteria which shall include such items as maximum allowable percentage of false reports and maximum allowable percentage of missed failure. The subcontractor shall provide the weight and the power consumption of Ethernet Switch.

The subcontractor must provide Monthly Report that contains following:

- Work that was performed by the subcontractor for the month covered by the report
- Work that will be performed by the subcontractor for the coming month

3.6 Design Information

The Subcontractor shall provide all necessary documents and deliverables for Hyundai-Rotem according to the time schedule defined by Hyundai-Rotem. The Subcontractor shall provide the interface information, which possibly affects performance, fitting, and form, from Hyundai-Rotem. The documents and the time schedule are listed and specified in part II of this specification "Design Information".

The Subcontractor shall provide the technical requirement and design information. (refer to section 4 and 5 of this PTS)

All drawing shall be provided by Auto-cad files and 3D Modeling files (STEP, IGES or CATIA V5).

3.7 Testing and Inspection

The Subcontractor shall be responsible for performance tests of the Ethernet Switch. The subcontractor shall be responsibility for a compressive series to tests to be performed to verify the suitability of

design and workmanship of each car according to the requirement in DENVER RTD TS requirement (Contract Part I, Section 4). And the subcontractor shall provide the test procedure and the test report after perform the each test. Hyundai Rotem and/or DENVER RTD have the right to witness any of these tests at the any stage of test progress.

Hyundai Rotem and/or DENVER RTD will carry out the First Article Inspection (FAI) on complete Ethernet Switch under the Subcontractor's responsibility. The Subcontractor shall carry out the type test and routine test of equipment and assembly. The commissioning test of the complete vehicle will be performed by Hyundai Rotem under the Subcontractor's responsibility for the Ethernet Switch and coupling assembly. The Subcontractor shall provide the required information for testing and carry out the tests and retrofit problems during the tests in accordance with PTS.

In the event that any test for the Ethernet Switch is failed, the Subcontractor shall, at his own expense, take whatever action is deemed such as, rectification, readjustment or design changes to the satisfaction of Rotem and DENVER RTD, in order to meet the testing requirements.

All the provided products will be required to satisfy the Material_Workmanship (refer to the Attachment of Denver_TP section 20).

The subcontractor must submit test report for every test and it includes but not limited to Pre-Production Test, First Article Inspection, Product Performance Test, Pre-Delivery Test and On-site Pilot Car Test.

3.8 Operation & Maintenance Manual

The Subcontractor shall provide necessary requirements for Operation and Maintenance Manual and Spare Parts Catalogues for the Ethernet Switch and special tool with hard copies and electronic format.

The requirement for Operating and Maintenance Manuals and Spare Parts Catalogues shall be provided for Approval by Hyundai Rotem according to the time schedule approved by Hyundai Rotem. The Subcontractor shall provide softcopy for all O&M Manual according to detail requirement (Attachment 10) and required formats provided from Hyundai Rotem.

The subcontractor shall provide the operation and maintenance manuals in accordance with Attachment of Denver_TP section 23.

3.9 Training

The Subcontractor shall provide comprehensive training to the DENVER RTD Employer's staff (maintenance, operating, training instructor and engineering) in accordance with DENVER RTD TS. The training activities and works for the Ethernet Switch shall be approved by Hyundai Rotem. The subcontractor shall prepare the training manual and provide it. The training manual shall be complied with the requirements of Hyundai Rotem.

The subcontractor shall provide training (document & information) in accordance with TP23 Attachment of Denver_TP section 23.

3.10 Warranty

The subcontractor has sole responsibility under this Contract to maintain sufficient warranty spare parts in this stock to support the warranty period requirements, according to the DENVER RTD Contract Part I, Section 5 or supply of equipment manufactured specifically for DENVER RTD.

4. Technical Specification

The Subcontractor, who will supply the Ethernet Switch for Denver project, shall comply with all sections related with Ethernet Switch of Technical Specification for the purchase of Electrical Multiple Unit Commuter Railcars (TS).

4.1 Technical Requirements

The Ethernet Switch can meet all technical requirement required in DENVER Technical Specification (TS). The subcontractor shall review the below section of TS, as a minimum, and comply;

1) Section 3.2 of Contract Part I
2) Section 3.3 of Contract Part I
3) Section 3.4 of Contract Part I
4) Section 3.6 of Contract Part I
5) Section 3.7 of Contract Part I
6) Section 3.8 of Contract Part I
7) Section 3.17 of Contract Part I
8) Section 3.19 of Contract Part I
9) Section 3.21 of Contract Part I
10) Section 4 of Contract Part I
11) Section 5 of Contract Part I
12) Section 9 of Contract Part I
13) Section 10 of Contract Part I
14) Section 11 of Contract Part I
15) Section 2 of Contract Part III
16) Section 12 of Attachment 7, Part C
17) Section 13 of Attachment 7, Part C
18) Section 14 of Attachment 7, Part C
19) Section 15 of Attachment 7, Part C
20) Section 16 of Attachment 7, Part C

21) TP Section 20

22) TP Section 23

4.2 Electromagnetic Compatibility

The subcontractor shall ensure that the subsystem and components operate correctly and reliably within the Line operating electromagnetic environment, and shall not themselves materially and adversely affect adjacent third party infrastructure and systems. The subsystem shall be electromagnetically compatible in the intended railway environment and with adjacent installations.

The Subcontractor shall comply with all relevant standards and legislation, including the following standard.

- IEC 62236-3-2 Railway applications - Electromagnetic compatibility

4.2.1 EMC Control

The Contractor shall submit electromagnetic compatibility control and test plan for approval and the plan shall insure that proper emphasis will be placed on the control of interference and adequate attention given to interface and Federal Communications Commission (FCC).

The subsystem shall operate without either suffering or causing interferences which may impact the system operation and/or safety because of conducted and/or radiated emissions and the contractor shall employ design techniques, construction methods, and whatever equipment is required to prevent interference caused by internal sources from affecting the proper operation of the car subsystem. In addition to coordinating frequencies, the Contractor shall provide necessary on-board balancing, filtering, shielding, modulating techniques and isolation to maintain signal to noise (S/N) ratios above limits. Shielding, isolating, balancing and grounding shall be used as required to reduce the undesirable effect of interference. Electrostatic and magnetic shielding methods shall be employed to minimize the effect of stray signals and transient voltages on interconnecting cables. Interconnecting power and signal cables shall be physically separated.

Cellular (wireless) telephones, typical farecards, credit/debit cards, laptop computers, PDAs and other magnetic storage media such as floppy disks shall not be adversely affected by EMI fields at any location inside and outside the vehicle, including the floor, ceiling, Operator's cab, or in proximity of equipment cabinets. For farecards, an exposure limit of 500 Gauss shall be considered as the standard for farecard susceptibility.

The Contractor shall ensure that any static or alternating fields generated by the subsystem shall not interfere with the correct operation of cardiac pacemakers, customer goods or any magnetic media. At a minimum the magnetic flux density in any car due to the subsystem shall not exceed the ICNIRP guideline elsewhere under any mode of operation, including failure modes.

4.2.2 Emission Limitations

The Contractor shall ensure that the cars do not produce any radiated emissions that will violate FCC Rules and Regulations, 47 CFR Part 15.103. In addition, for purposes of design validation, the subsystem shall not produce radiated emission levels above those permitted by Figure 10-2(UM05), Curve 2 of MIL-STD-461E when measured 100 feet from the track centerline. The radiated emissions of the Pilot Car shall not exceed the level permitted by the above standard at any frequency between 0.15 MHz and 1000MHz. Testing shall be based on the document "Radiated Interference in Rapid Transit Systems, Volume 2 - Suggested Test Procedures, UMTA-MA-06-0153-85-11" for guidance and shall be conducted to the satisfaction of the Engineer.

4.2.3 Safety Analysis

The Contractor shall prepare an EMI safety analysis which demonstrates the protection against hazards due to EMI. A draft analysis shall be submitted for review at the early Preliminary Design Review; an updated version shall be submitted at the mid-point in the design process between the Preliminary Design Review and the Final Design Review and the completed analysis shall be submitted at the early Final Design Review.

The Contractor shall document through the EMI safety analysis that the subsystem's EMI emissions are safe under normal conditions, and that the subsystem provides adequate detection, annunciation, and response to failures which could cause EMI to increase above acceptable levels. The EMI safety analysis shall include a Preliminary Hazard Analysis; a Fault Tree; and a Failure Mode Effects Analysis.

The EMI safety analysis shall document the protective actions which prevent a hazard occurrence. The EMI safety analysis shall demonstrate that the equipment and its actions are adequate to prevent an EMI hazard. The EMI safety analysis shall distinguish between failures which affect EMI and for which there is automatic protection, those which are annunciated for Operator or Maintainer action, and those which are undetected failures.

The EMI safety analysis shall consider all relevant failures of the subsystem, including failures of line filter components, EMI detection and annunciation systems, and any other components that can affect the level or frequency of the generated EMI; unusual operating conditions; and failures of wayside equipment affecting, such as a substation component failure. The EMI safety analysis must also include equipment self-testing in areas that affect EMI.

4.2.4 Compatibility Design Report

The Contractor shall prepare an Electromagnetic Compatibility (EMC) Design Report prior to Final Design Review, and subsequently update it when and if design changes occur as a result of testing. The report shall consist of two parts, Emissions Report and Susceptibility Report, as described below.

The Emissions Report shall document:

a) EMC emission control methods, including equipment layout, circuit routing, frequency coordination and stability, filters, interconnections, grounding, and shielding. The Contractor shall

comply with IEEE Standard 1100-1992, Recommended Practice for Powering and Grounding Sensitive Electronic Equipment, unless otherwise approved;

b) Description of each unit with a peak power rating of greater than 5 kW, including the emission characteristics versus operating mode, voltage and loading, as well as line filter, output filter, line input impedance, and circuit and operating mode considerations;

c) Characteristics of any repeating narrow band emissions and frequency stability control measures.

The Susceptibility Report shall document:

a) EMC susceptibility control methods, including equipment layout, equipment enclosure design, circuit routing, frequency coordination and stability, filters, interconnections, grounding, and shielding;

b) The high voltage and low voltage grounding arrangement, and provisions for worst-case ground or return bus voltages within the car;

d) The normal and fault paths for ground currents;

e) The equipment protection as well as any restrictions on equipment operation or maintenance necessary to ensure susceptibility control, such as not leaving covers off of equipment enclosures; and

f) For each type of electrical interface between subsystems or enclosures:

i) The driving circuit, including impedance and maximum and minimum output voltage and current levels;

ii) The receiver circuit, including maximum and minimum voltage or current thresholds, margin against error, and destructive threshold;

iii) All aspects of the connection requirements, such as single wire, double wire, twisted pair, shield, and shield grounding; and

iv) Any special circuit or connection provisions needed to ensure EMC.

4.3 Fire

4.3.1 Fire Safety

The fire safety analysis for each equipment shall be submitted in accordance with all provisions of 49 CFR 238.103 (c), Fire Safety Analysis for Procuring New Passenger Equipment, and APTA RP-PS-005-00, Fire Safety Analysis of Existing Passenger Rail Equipment.

4.3.2 Flammability, Smoke Emission and Toxicity Requirement

The sub-supplier shall submit a list of combustible materials intended to be used in the Car. The sub-supplier shall test combustible materials used in Cars to demonstrate compliance with the requirements set forth in 49 CFR 238.103 and NFPA 130.

The sub-supplier shall submit fire test reports from an independent laboratory indicating successful testing and demonstrating compliance with such requirements. Testing shall be

conducted after the Phase 1 Effective Date and on a production batch of material intended to be used on the Car. Each laboratory shall have tested a standard test sample no greater than 30 days prior to performing the tests, the results of which shall be submitted to Hyundai-Rotem.

Materials used in Car construction, except for materials used in small parts such as knobs, rollers, fasteners, clips, grommets, and small electrical parts that would not contribute significantly to fire propagation or to smoke or toxic gas generation and are distributed throughout the Car, shall be tested for toxicity using Boeing Specification Support Standard BSS-7239. Materials shall comply with the following maximum toxic gas release limits (ppm):

Carbon Monoxide (CO)	3,500 ppm
Hydrogen Fluoride (HF)	200 ppm
Nitrogen Dioxide (NO2)	100 ppm
Hydrogen Chloride (HCL)	500 ppm
Hydrogen Cyanide (HCN)	150 ppm
Sulfur Dioxide (SO2)	100 ppm

Tests shall be conducted in flaming mode after 240 seconds using a National Bureau of Standards smoke density chamber for sample combustion. Gas sampling may be conducted during the smoke density test.

All fire test results for the material used to DENVER project must be submitted. The previous fire test of other project results cannot be allowable.

4.3.3 Electrical Fire Safety

Electrical equipment, wiring and apparatus shall conform to the standards of NFPA 130, Section 8.

4.3.4 Fire Load Calculation

Fire load calculation for all combustible material must be calculated with heat value tested in accordance with ASTM E 1354.

4.3.5 Fire Performance Deliverables

The Fire Performance Deliverables shall be provided with the following documents:

No	Submission Document	Remark
1	Fire Safety Analysis for each equipment	Reference standard - 49 CFR 238.103 (c) and APTA RP-PS-005-00
2	Report for flammability, Smoke Emission and Toxicity	The summary table for FST test shall be delivered with test report. Refer to the attached combustible material matrix

No	Submission Document	Remark
3	Combustible material matrix	Refer to the attached combustible material matrix
4	Heat Release Rate test report	Report tested in accordance with ASTM E 1354
5	Description of design in specific areas of fire risk	

*Attachment Combustible Material Matrix

Combustible Material Matrix

4.4 RAMS

4.4.1 Reliability Requirement

The Subcontractor shall meet the specified reliability targets with annual design mileage of 125,000 miles per Car

Equipment	MDBSF (car mile)	MDBF (car mile)
Ethernet switch	Not Applicable	Not Applicable

Table 1: Reliability Targets

For failure definition, please refer to Section 5 in End User Specification Part I Scope of Work.

4.4.2 Fleet Defects (Pattern Failures)

The occurrence of independent failures of the same warranted item that exceeds more than 10 percent of the total number of identical items supplied under the subcontract may be declared a fleet defect or pattern failure. On this basis, the subcontractor shall be required to develop and implement a corrective action program to eliminate the pattern failure. The corrective action program must be approved by Rotem/End User and its implementation must be coordinated with Rotem.

4.4.3 Maintainability Requirement

I. Subcontract shall propose and demonstrate MTTR (Mean Time To Repair) based on the previous project through maintainability analysis.

II. The equipment shall be designed and constructed to have a minimum period of 92 days between periodic maintenance servicing by shop personnel.

III. The Schedule Preventive Maintenance Plan will include a designation of inspection intervals conformant with all FRA requirements, and include the inspection and maintenance standards, service intervals, wear limits, settings and tolerances.

IV. Each maintenance task shall be defined in the 'Scheduled Preventive Maintenance Plan' in the format provided by Hyundai-Rotem. The 'Scheduled Preventive Maintenance Plan' shall be in compliance with the maintenance requirements specified in the Maintenance Manuals prepared by the supplier. The 'Scheduled Preventive Maintenance Plan' shall provide all information necessary for identification and quantification of the required maintenance resources.(Refer to Appendix IV)

V. Each maintenance task shall be followed by reference numbers on the maintenance manual for Hyunai-Rotem's review.

VI. At hte supplier's cost, in RTP's Depot, in coordination with Hyundai-Rotem, the supplier shall demonstrate the maintainability. (Part1_5.3.Maintainability). The procedures used in the demonstration shall be the same as those included in the manuals delivered.

VII. With the exception of consumables (motor brushes, brake shoes etc), overhaul period of each equipment shall be decided considering operating hours, cycles and running distance.

VIII. Items without a specified maintenance or overhaul period shall be designed not to require replacement during the vehicle design life.

4.4.4 Regulation and Standard Requirement

Subcontractors shall comply with 49 CFR Part 238.105, APTA's manual, MIL-STD-882C, MIL-STD-1629 and CENELEC standard EN50128 specified in attached file.

RAMS approach.pdf

4.4.5 Safety Requirement

I. The Subcontractor shall ensure that the Risk Index of all hazards on the train shall not be worse than the "Acceptable" category i.e. either "Ar" or Ac" as defined in Appendix II of attached RAMS guideline.

II. The subcontractor shall comply with the general safety design requirement specified in Section 4.3 of Mil-STD-882C.

III. Hazards of Category I or II severity shall be demonstrated by test, the validity of any portion of all analyses.

4.4.6 Failure Analysis

In the event of a failure of any component during contractually required testing or the warranty

period, the subcontractor shall submit a complete failure analysis report to Rotem for approval. The failure analysis shall be submitted within 30 days of receipt of the failed part by Subcontractor, unless otherwise agreed upon. Unless otherwise specified, the format of failure report shall be mutually agreed.

4.4.7 Deliverables

I. Subcontractor shall submit Hardware and Software Safety Program and its supporting analysis to comply with 49CFR 238.105 for all micro-based safety related systems by within 2 weeks after order by Rotem and it shall include the contents specified in Section 4.3 in 49CRF 238.105

II. Subcontractor have to submit the Safety Integrity Level Justification Document in order to justify the appropriate SIL rating and for approval.

III. The System Hierarchy including LRU (Line Replaceable Unit) and LLRU (Lowest Line Replaceable Unit) list using the attached table by within 2 weeks after order by Rotem.(Refer to Appendix I)

IV. The subcontractor shall prepare and submit Reliability and Maintainability Analysis Report at Preliminary Design Review and Final Design Review stage and the contents of it shall consist of, but not be limited to, the following elements

- Definition of Reliability and Maintainability goals and criteria
- Subsystem functional description including subsystem schematics and functional flow diagrams
- A summary of reliability and maintainability prediction results and remedial actions taken as a result of the Reliability and Maintainability Analysis
- A comparison of reliability and maintainability predictions against allocated requirements specified in Section 1.1 and 1.2
- The format for reliability analysis, Maintainability analysis shall be provided in Appendix III, IV and V.

V. The subcontractor shall prepare and submit Failure Modes and Effects and Criticality Analysis at Preliminary Design Review and Final Design Review stage

VI. Fault Tree Analysis including source data shall be submitted at Preliminary Design Review and Final Design Review stage.

The Hazards of Category I or II severity shall be subject to Fault Tree Analysis.

VII. The subcontractor shall prepare and submit Safety Analysis Report at Preliminary Design Review and Final Design Review stage and the contents of it shall consist of, but not be limited to, the following elements.

- Definition of safety goals and criteria
- Subsystem functional description including subsystem schematics and functional flow diagrams
- Description of Safety Feature

- A summary of the objectives, findings and remedial actions taken as a result of the Safety Analysis
- Hazard Analysis including Preliminary Hazard Analysis (PHA), System Hazard Analysis if identified, Sub-system Hazard Analysis, Operating and Support Hazard Analysis using attached format(Appendix VII, VIII, IX and X)

VIII. The subcontractor shall provide information regarding Safety and Security Certification and Threat and Vulnerability Analysis and resolution (Refer to End User Specification_ Attachment 9 Project and Construction Management)

4.5 Software

4.5.1 Software Requirement

The following requirement shall be applied and satisfied.

A. Subcontractor must comply with the section 42 of Contract Part III(General Terms-Standard) for Intellectual Property Right.

B. Comply with section 3.21.3 of Contract Part I(Scope of Work) and 16.3 of Attachment 7 to the Concession Agreement.

C. The following CDRL specified in Attachment 6 of Concession Agreement shall be submitted

- CA-093(Source Code Escrow Agreement), 7B-16(Software Management Control Plan), 9-08(Software Requirement Manual)

D. Comply with 9.3.3 of Attachment 7 of Concession Agreement

E. Comply with 2.5(c) of Attachment 9 of Concession Agreement

All software supplied for the system designed structurally and logically, fully documented, thoroughly tested in a systematic manner such that they can achieve high quality in terms of safety, reliability, testability, traceability and maintainability. Also all software supplied for the system shall be developed in line with latest ISO 90003

Subcontractor has to comply with Rotem's Software Quality Assurance Plan which will be approved by Customer. Software development follows FRA 49CFR236 Subpart I requirements as the primary document. Rotem's Software Quality Assurance Plan can be used as a reference. In case of conflict between documents, Wabtec will follow FRA requirements.

Subcontractor shall provide the properties and all requested material for Rotem's software quality & safety audit to Subcontractor. Corrective action against NCR and OBS should be returned to Rotem within one week after software audit. And also, Implementation of corrective actions against to Identified NCR (Non conformance Report) and OBS (Observations) should be finalized within one-month after software audit. Rotem will not pay the amount of money for software development without the completion of the identified NCR and OBS. Otherwise, Subcontractor has to keep all relevant quality record (i.e. internal software quality audit report, review record etc.) for Hyundai Rotem's software quality audit. The systems as proposed are based upon existing software, with modifications to meet DTP and Rotem application requirements. Wabtec retains Intellectual Property rights for all delivered software. Any software requiring escrow by Rotem will require a software non-disclosure agreement.

HR

The Subcontractor shall be obliged to take care of any software change if Rotem and Customer's

request the change for the correction of software during commissioning phase, commissioning and Warranty Period, etc. The change shall be implemented by the Subcontractor. In the event, when any non-conformity arises to the specified requirements, the Subcontractor shall take remedial measures at its own cost within the schedule to be agreed with Rotem.

The Subcontractor shall be obliged to abide by the software change control procedure on site (or online based change control system (http://sccp.rotem.co.kr) which should be prepared by Rotem in order to enhance the regulation of software change in train-set. It should be not allowed to upload the changed software into train-set without the approval by Rotem/Customer under software change control by Rotem. Subcontractor has to supply test report of system integration testing or bench testing of the updated software performed subcontractor's premises before formal release of the updated software.

During Test & commissioning stage, subcontract must inform the software version number and it's revision history document on the monthly basis.

The Subcontractor shall provide Diagnostic & Maintenance Software and its user manual.

The Subcontractor has to supply back-up copies of software for uploading purpose whenever software change is occurred.

4.5.2 Details of Escrow Requirement

Regarding the requirement of supply of source code and work product specified in section 41.2.4 and 42.1.8 of Contract Part III(General Terms – Standard), the following requirement must be applied for escrow agreement and establishment. But, below could be subject to the discussion with customer and/or end-user.

Subcontractor as software owner is totally responsible for setup and completion of software escrow agreement with Rotem's customer. The escrow period should be followed (e.g. over 35 years) by the original contract between DTP and RTD.

♣ Fee for Escrow Service

E&M contractor is in charge of escrow agent fee including setup fee, annual fees. However, the cost of license and/or activities for the preparation of escrow package (including the license of software development tool) and activities for escrow verification testing shall be born by subcontractor.

If the update of Escrow package after deposit is needed due to software failure, the fee for updating shall be born by subcontractor.

Also, escrow agent can perform escrow verification testing. In that case, E&M Contractor is in charge of fees for escrow Verification Testing. However, the cost for preparation of this escrow verification testing shall be born by subcontractor.

♣ Selection of Escrow Agent

Supplier shall comply with end-customer's preferring escrow agent (e.g. Iron Mountain) and comply with end-customer's presented escrow agreement document based on the selected escrow agent.

We can't accept supplier's preferring escrow agent and proprietary escrow agreement document in any case.

♣ Escrow Material List

The Escrow material to be deposited to Escrow Agent shall contain mandatorily as follows;

- Source code and related data file.

- Complier, link, design tool, proprietary tool etc. to generate application software from source code(must include of license information and transfer license to end user if certain terms and condition is reached) at supplier's expenses.

- Guideline document(build procedure) to generate application software from source code.

- Software Technical Document such as software design description and software requirement specification etc.

- Properly documented (architecture and planning, library software, operational system and settings, communication protocols, files, etc)

- Others if requested by end-user

♣ Escrow Verification Testing

The Escrow Verification Testing Plan shall be submitted by subcontractor within 30 days before the testing.

For escrow verification testing, subcontractor has to have system or simulator for identifying software version number and performing function testing.

The Escrow Verification Testing Plan shall contain contents as follows;

The supplier shall identify:

A. where the verification is to take place

B. the personnel required

C. the expected duration of the verification exercise. Time must be included to verify the Escrow deliverables match the items identified in the Escrow Materials List

D. the procedures to be used to build the software (on of the items required in the Escrow Material List). These procedures will be followed to create the executables

E. the method to be used to verify that the information assembled results in the software executables are same to active software in trainset. This will depend on the software supplied and should be;

- a comparison of checksums and program size and/or
- a comparison of software version number after installation the compiled software into the system
- execution of a defined set of tests to prove correct operation of the software
- execution of a 'file compare' program or equivalent.

Otherwise, escrow agent can perform escrow verification testing as per customer's request in compliance with customer's escrow agreement document. Supplier must fully support escrow agent to perform escrow verification testing if requested without any cost implications.

4.5.3 Software Documentation

The following software documents based on IEEE-STD-1558 (Type 5) shall be submitted. For individual software documents, the specific IEEE standard should be applied.

- Software Project Management Plan
- Software Quality Assurance Plan
- Software Configuration Management Plan
- Software Verification and Validation Plan
- Software Verification and Validation Report
- Software Requirement Specification
- Interface Control Document
- Software Design Description
- Software Requirements Specification
- Software Requirement Traceability Matrix
- Software Test Plan
- Software Test Procedure
- Software Test Report
- Software Version Description Document
- Software User Manual
- Software Requirement Manual(CDRL #9-08)

4.6 Noise

Not Applicable

5. Design Information

In order to take an approval for the Ethernet Switch from DENVER, the subcontractor shall provide all technical information as a written document. The below document/deliverable shall be provided as a minimum.

All technical information shall follow the “publications style guide” attached in TS.

5.1 General

The Subcontractor shall submit, the following general information in accordance with the schedule:

No	Document / Deliverables	Reference	*Deadline
1	Test and Inspection plan		At PDR
2	As-built drawing		At PDR

3	Operation & Maintenance Manual		At PDR
4	Illustrated parts catalog		At PDR
5	Training plan		At PDR
6	Training material		At PDR
7	List of warranty period spare parts		At PDR
8	List of recommended spare part		At PDR
9	List of special tools and test equipment		At PDR
10	PCB information		At PDR
11	Software Source code		At PDR

* Note: The deadline to be discussed and agreed between the Subcontractor and Hyundai-Rotem.

5.2 Design

The Subcontractor shall submit, the following design information in accordance with the schedule:

No	Document / Deliverables	Reference	*Deadline
1	Clause by Clause Compliance sheet for the Technical Specification of DENVER		Before Contraction
2	Dimensional Assembly Drawing by Autocad files and 3D modeling files		At CDR
3	Sub-Assembly Drawing		At CDR
4	System description		At CDR
5	Function description		At CDR
6	Product description		At CDR
7	System Block Diagram or Schematic Diagram		At CDR
8	Installation Specification of On-board equipment		At CDR
9	Installation Specification and investigation report of Wayside equipment		At CDR
10	Interface Diagram and Specification with Communications, CCTV and PTC		At CDR
11	Interface Diagram and Specification with Wayside Equipment		At CDR
12	Cabling requirement		At CDR
13	Test Specification		At PDR

14	Test Report		At PDR
15	Test Specification of Ethernet Switch at Complete Vehicle		At PDR
16	Test Report of Complete Vehicle		At PDR
17	Commissioning Test Procedure for Ethernet Switch (Including interface test with Wayside Equipment)		At PDR
18	Test Report of Commissioning Test		At PDR

* Note: The deadline to be discussed and agreed between the Subcontractor and Hyundai-Rotem

5.3 SEMS

The subcontractor shall provide all design information requested in the clause 4 of this PTS for EMI/EMC, Fire, RAMS and Software.

6. Tests and Inspection

The subcontractor shall perform all tests specified in the TS. The test procedure and the report shall be prepared and provided by the subcontractor.

7. Attachment

DENVER RTD EMU_ TS

Part I – Scope of Work.pdf

Part III – General Terms – Standard

Attachment 7.pdf

DENVER_TP Section 20.pdf

DENVER_TP Section 23.pdf

DENVER RTD SEMS REQUIREMENT

Attachment_SEMS Requirement.zip